

Arman Dinarvand

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EDUCATION

BSc in Animal Biology

Kharazmi University

09/2022 – Expected: 06/2026

RELEVANT COURSEWORK

Animal Physiology II: Neural and Endocrine Systems, Animal Behaviors, Deep Learning, Neural Dynamics, Differential Equations, Fundamentals of Neuroscience, Fundamentals of Statistics, Mathematics for Machine Learning, Bioinformatics, Control Theory, Computational Neuroscience, Information Theory, Computer Vision

RESEARCH EXPERIENCE

- **Lead Researcher** Jun 2025 – Present
Temporal and Spatial Dynamics of Atrophy in Alzheimer's Disease
Department of Animal Biology, Kharazmi University
- **Research Assistant** Mar 2025 – Dec 2025
CeRNA Network Dysregulation in Autism Spectrum Disorder
Department of Cell and Molecular Biology, Kharazmi University
- **Research Assistant** Nov 2024 – Apr 2025
Introduction of Novel Anti-Angiogenic VEGFR2 Inhibitors Using Machine Learning and Deep Learning
Department of Cell and Molecular Biology, Kharazmi University

TEACHING EXPERIENCE

- **Teaching Assistant, Bioinformatics** 2024
Department of Cell and Molecular Biology, Kharazmi University

PUBLICATIONS

- **Arman, D.** (2026). neurocomplexity: a Python toolkit for criticality, information flow, and dimensionality analyses of neural populations. *Manuscript in preparation*.
Developed an open-source object-oriented Python package to quantify complex systems properties in high-dimensional neural time-series data. Implemented core analytical frameworks for thermodynamic criticality (branching ratios, scaling exponents), information dynamics (transfer entropy, active information storage via JIDT/IDTxL wrappers), and state-space geometry using advanced dimensionality reduction (UMAP, t-SNE, and custom topological tools). Optimized for seamless integration with Neurodata Without Borders (NWB) and standard spiking/LFP datasets.
- Karami, L., Heidari, S., & **Arman, D.** (2026). Screening and Classification of Anti-Angiogenic VEGFR2 Inhibitors with Supervised Machine Learning, Deep Learning, Molecular Docking and Molecular Dynamics Simulation. *Journal of Molecular Graphics and Modelling*. DOI: doi.org/10.1016/j.jmgm.2026.109397
My work involved the end-to-end pipeline, from developing predictive DL models for virtual screening to conducting in-silico ADMET analysis and molecular docking to evaluate drug-likeness and binding affinity. A key contribution was troubleshooting and optimizing the deployment of

the final machine learning models for robust inference.

- **Arman, D.**, Moradniaei, A., Amini, E., Borna, K. (2026). Temporal Dynamics of Atrophy in Alzheimer’s Disease Within the Executive Control Network: A Spatio-Temporal Graph Attention Network Approach. *Manuscript in preparation.*

Led a research project to model the spatio-temporal sequence of cortical atrophy in Alzheimer’s Disease using a Spatio-Temporal Graph Attention Network (ST-GAT). My responsibilities encompassed end-to-end project leadership, including study design, image preprocessing, model development and formal analysis, and the composition of the initial research manuscript.

- **Arman, D.**, Doroudian, M. (2026). CeRNA Network Dysregulation in Autism Spectrum Disorder. *Under Review.*

Conducted a comprehensive review of ceRNA network dysregulation, with a focus on systemic mechanisms contributing to the etiology of autism spectrum disorder.

RESEARCH SKILLS

- **Computational Neuroscience & Biophysical Modeling:** Spiking Neural Networks (SNNs), Conductance-based Single-Neuron Models, Cortical Microcircuitry Simulations, Drift-Diffusion Models (DDMs), Network Topology & Graph Theory. (*Tools: NEURON, Brian2, NetPyNE, NetLogo*)
- **Deep Learning & Geometric ML:** Wasserstein GANs, Spatio-Temporal Graph Attention Networks (ST-GAT), Recurrent Neural Networks (RNN/LSTM), Transformers, State-Space Models, Neural Latent Factor Analysis (LFADS, GPFA), Dimensionality Reduction (UMAP, t-SNE, PCA). (*Tools: PyTorch, TensorFlow*)
- **Neural Electrophysiology & Signal Processing:** Extracellular Recording Analysis (Single-unit Spike Sorting, LFP Analytics), EEG Processing, Time-Frequency Analytics, Spectral Decomposition, Neural Decoding/Encoding (Rate, Temporal, Sparse, Efficient Coding). (*Tools: Kilosort, Phy, Elephant, MNE-Python, NWB format*)
- **Neuroimaging & Brain Mapping:** Structural & Functional Connectivity Networks, Cortical Atrophy Mapping, Automated Tissue Segmentation, Morphometric Voxel Analysis. (*Tools: FSL, Plotly*)
- **Theoretical & Complex Systems Frameworks:** Complex Systems Theory (Criticality, Autonomy), Information Dynamics (Transfer Entropy, Active Information Storage), Dynamical Systems Theory (Attractor Landscapes, Bifurcation, Phase Stability), Control Theory, Bayesian Inference, Change-Point Probability, Hebbian Learning. (*Tools: pyDSTool*)
- **Wet Lab & Molecular Biology:** Mammalian Cell Culture, Polymerase Chain Reaction (PCR), DNA Extraction, Gel Electrophoresis, High-Dimensional Molecular Screening, Bioinformatic Molecular Simulations.

LANGUAGES & IT SKILLS

- **Languages:** Persian (Native), English (C1), French (Basics)
- **Programming:** Python, MATLAB, R, Git
- **Tools & Frameworks:** PyTorch, TensorFlow, NEURON, Brian2, NetPyNE, DeepLabCut, GPFA/LFADS, Kilosort, Plotly, pyDSTool, UMAP/t-SNE, FSL, Elephant, Phy, NWB, NetLogo, FreeSurfer, IDTxl

SELECTED TALKS

- **Workshop:** “Introduction to Neural Dynamics and AI” Mar 2025
Kharazmi University

REFEREES

- **Dr. Mohammad Doroudian**
Assistant Professor, Department of Cell and Molecular Biology
Kharazmi University, Tehran, Iran
mdoroudi@tcd.ie
- **Dr. Elaheh Amini**
Assistant Professor, Department of Animal Biology
Kharazmi University, Tehran, Iran
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- **Dr. Leila Karami**
Assistant Professor, Department of Cell and Molecular Biology
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